



Security Operations Center (SOC) Fundamentals

1. Introduction

A Security Operations Center (SOC) is responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents. This report demonstrates an understanding of alert prioritization, incident classification, incident response procedures, evidence preservation, and security monitoring activities. The objective of this exercise is to develop practical SOC analyst skills and understand the complete incident handling lifecycle.

2. Alert Priority Levels

Security alerts are prioritized according to their severity, impact, and urgency.

Critical

Critical alerts indicate severe security incidents requiring immediate action. These incidents may cause significant business disruption or compromise sensitive information.

Examples:

- Ransomware attack
- Active data breach
- Remote code execution

High

High-priority alerts indicate serious threats that require prompt investigation.

Examples:

- Unauthorized administrator access
- Malware infection
- Privilege escalation

Medium

Medium-priority alerts represent suspicious activities that may require investigation.

Examples:

- Multiple failed login attempts
- Brute-force attacks

- Unusual network traffic

Low

Low-priority alerts indicate minor security events with limited impact.

Examples:

- Port scanning
- Routine policy violations
- Informational events

CVSS Scoring

The Common Vulnerability Scoring System (CVSS) measures the severity of vulnerabilities on a scale from 0 to 10.

Score Range	Severity
0.0 – 3.9	Low
4.0 – 6.9	Medium
7.0 – 8.9	High
9.0 – 10.0	Critical

3. Incident Classification

Security incidents are categorized according to their nature and impact.

Malware

Malicious software designed to damage systems or steal information.

Examples:

- Virus
- Worm
- Trojan
- Ransomware

Phishing

Fraudulent attempts to obtain sensitive information through deceptive emails or websites.



Distributed Denial of Service (DDoS)

An attack that overwhelms systems with traffic, causing service disruption.

Insider Threat

A threat originating from individuals with authorized access to organizational resources.

Data Exfiltration

Unauthorized transfer of sensitive information outside the organization.

MITRE ATT&CK Framework

MITRE ATT&CK is a knowledge base that documents attacker tactics and techniques used during cyberattacks.

VERIS Framework

VERIS provides a standardized method for recording and analyzing security incidents.

ENISA Taxonomy

ENISA Incident Taxonomy helps classify incidents consistently across organizations.

4. Basic Incident Response

The Incident Response Lifecycle consists of six phases.

Preparation

Develop policies, procedures, tools, and response plans before incidents occur.

Identification

Detect and verify potential security incidents.

Containment

Limit the spread and impact of the incident.

Eradication

Remove malicious artifacts and eliminate the root cause.



Recovery

Restore systems and services to normal operation.

Lessons Learned

Analyze the incident and improve future response capabilities.

5. Alert Management Practice

Alert Classification Table

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	Log4Shell Exploit	Critical	T1190
003	Port Scan	Low	T1046

The screenshot shows a Google Sheets interface with a spreadsheet titled 'Untitled spreadsheet - Google Sh...'. The spreadsheet contains the following data:

Alert ID	Alert Type	Priority	MITRE Technique
1	Phishing Email	High	T1566
2	Log4Shell Exploit	Critical	T1190
3	Port Scan	Low	T1046

Incident Ticket

Title:

[Critical] Ransomware Detected on Server-X

Description:

Ransomware activity was detected on Server-X. Investigation identified suspicious file execution and potential encryption activity.



Indicators of Compromise:

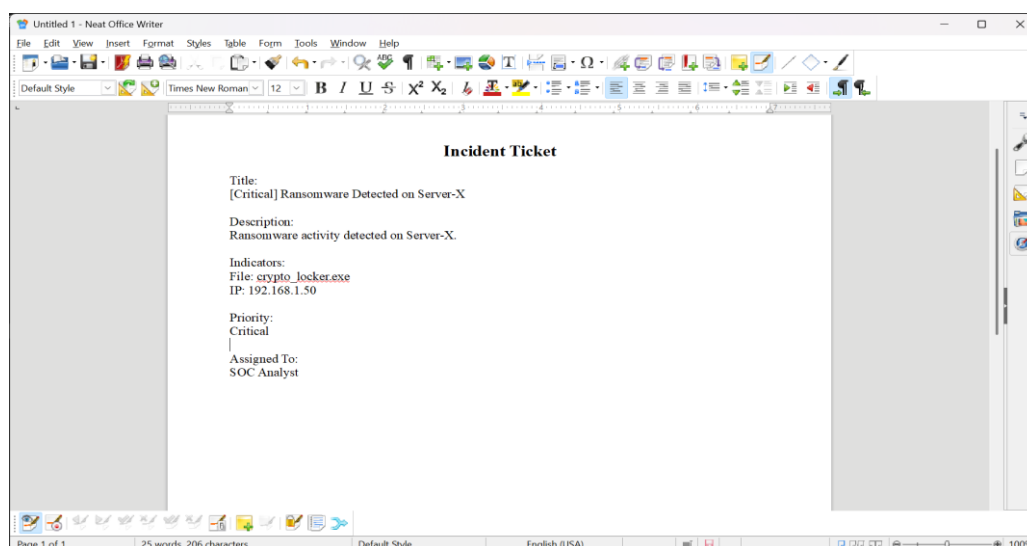
- File: crypto_locker.exe
- IP Address: 192.168.1.50

Priority:

Critical

Assigned To:

SOC Analyst



Escalation Email

Subject: Critical Security Incident Escalation

Dear Tier 2 Team,

A critical ransomware alert has been detected on Server-X. Initial investigation identified suspicious file activity associated with crypto_locker.exe and network communication from IP address 192.168.1.50.

The affected system has been isolated to prevent further impact. Immediate assistance is requested for advanced investigation and remediation.

Regards,

SOC Analyst



6. Response Documentation

Executive Summary

A phishing email containing a malicious link was detected and investigated. The incident was contained before significant impact occurred.

Timeline

Timestamp	Action
14:00	Endpoint isolated
14:30	Memory dump collected
15:00	IOC analysis completed

Impact Analysis

One workstation was affected. No evidence of data loss was identified.

Remediation Steps

- Removed malicious email
- Blocked malicious sender
- Reset user password
- Updated antivirus definitions

Lessons Learned

Additional user awareness training and email filtering controls should be implemented to reduce future phishing risks.

7. Alert Triage Practice

Alert Analysis

Alert ID	Description	Source IP	Priority	Status
002	SSH Brute Force	192.168.1.100	Medium	Open

Threat Intelligence Validation

The source IP address was analyzed using threat intelligence platforms. Reputation analysis was performed to identify any known malicious activity associated with the indicator.

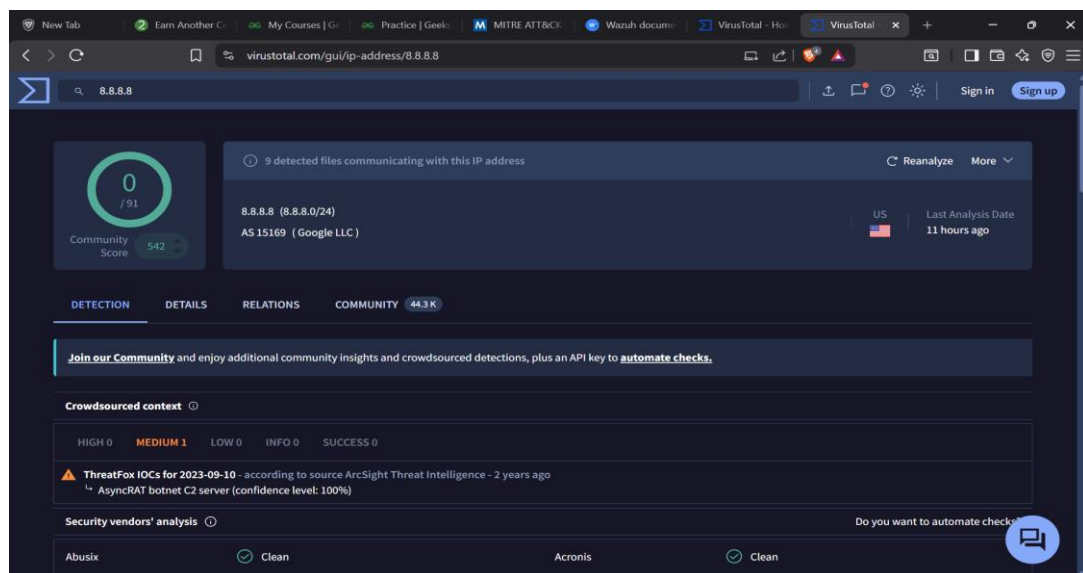


Result:

No immediate threat was confirmed; however, the activity was monitored for further investigation.

The screenshot shows a Google Sheets spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J
	Alert ID	Description	Source IP	Priority	Status					
1										
2	2	SSH Brute Force	192.168.1.100	Medium	Open					
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										



8. Evidence Preservation

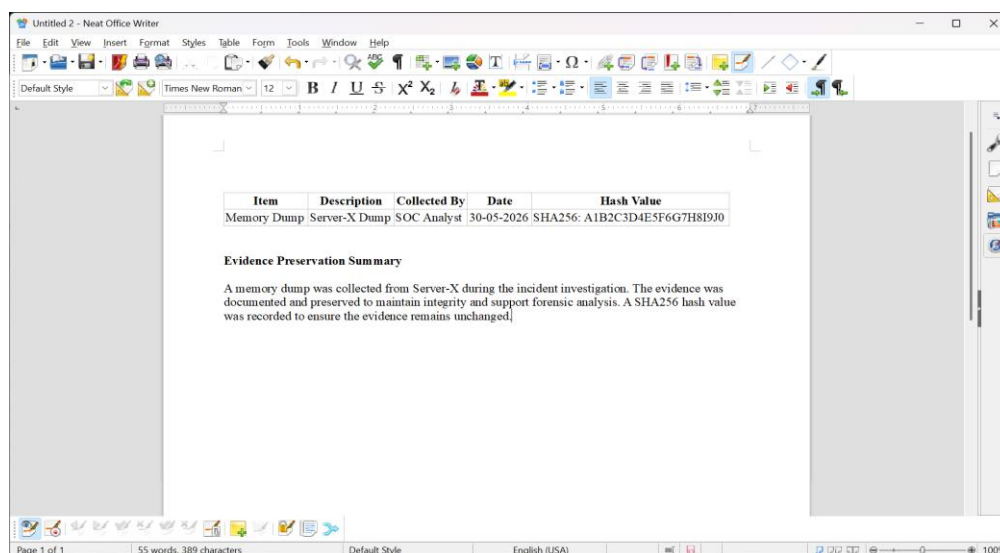
Evidence Collection Log



Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Dump	SOC Analyst	30-05-2026	SHA256: A1B2C3D4E5F6G7H8I9J0

Evidence Preservation Summary

A memory dump was collected from Server-X during the investigation. The evidence was documented and preserved according to forensic best practices. A SHA256 hash value was recorded to verify data integrity and ensure the evidence remains unchanged throughout the investigation process.



9. Capstone Project – Full Alert-to-Response Cycle

Attack Simulation

A vulnerable FTP service was targeted by an attacker attempting unauthorized access.

Detection

Security monitoring systems generated alerts indicating suspicious FTP activity. **Triage**

SOC analysts reviewed system logs and identified indicators consistent with exploitation attempts.



Containment

The affected system was isolated from the network to prevent further compromise.

Eradication

Malicious files and unauthorized access mechanisms were removed from the system.

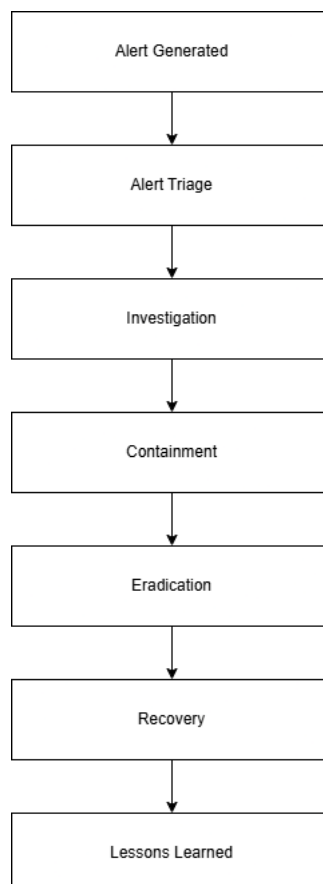
Recovery

The system was restored to normal operation and security patches were applied.

Recommendations

- Perform regular vulnerability assessments.
- Apply security patches promptly.
- Implement continuous monitoring.
- Conduct security awareness training.

Workflow Diagram





10. Conclusion

This Week 2 SOC assignment provided practical exposure to alert prioritization, incident classification, incident response procedures, alert triage, evidence preservation, and security operations workflows. The activities enhanced understanding of how SOC analysts detect, investigate, contain, and remediate cybersecurity incidents while maintaining proper documentation and evidence integrity.

11. References

1. MITRE ATT&CK Framework – <https://attack.mitre.org>
2. NIST SP 800-61 Incident Handling Guide – <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf>
3. FIRST CVSS Documentation – <https://www.first.org/cvss>
4. Wazuh Documentation – <https://documentation.wazuh.com>
5. TheHive Documentation – <https://thehive-project.org>
6. VirusTotal – <https://www.virustotal.com>
7. Draw.io – <https://app.diagrams.net>



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